Dirty money: an investigation into the hygiene status of some of the world's currencies as obtained from food outlets.

A total of 1280 banknotes were obtained from food outlets in 10 different countries (Australia, Burkina Faso, China, Ireland, the Netherlands, New Zealand, Nigeria, Mexico, the United Kingdom, and the United States), and their bacterial content was enumerated. The presence of bacteria on banknotes was found to be influenced by the material of the notes, and there was a strong correlation between the number of bacteria per square centimeter and a series of indicators of economic prosperity of the various countries. The strongest correlation was found with the "index of economic freedom," indicating that the lower the index value, the higher the typical bacterial content on the banknotes in circulation. Other factors that appear to influence the number of bacteria on banknotes were the age of the banknotes and the material used to produce the notes (polymer-based vs. cotton-based). The banknotes were also screened for the presence of a range of pathogens. It was found that pathogens could only be isolated after enrichment and their mere presence does not appear to be alarming. In light of our international findings, it is recommended that current guidelines as they apply in most countries with regard to the concurrent hygienic handling of foods and money should be universally adopted. This includes that, in some instances, the handling of food and money have to be physically separated by employing separate individuals to carry out one task each; whereas in other instances, it could be advantageous to handle food only with a gloved hand and money with the other hand. If neither of these precautions can be effectively implemented, it is highly recommended that food service personnel practice proper hand washing procedures after handling money and before handling food.